CCIT Report #381 May 2002

Theoretical Performance Analysis of the General Transfer Function GSC

Sharon Gannot, David Burshtein and Ehud Weinstein

S. Gannot: *Department of Electrical Engineering, Technion — Israel Institute of Technology, Technion City, Haifa 32000, Israel*

E-mail: gannot@siglab.technion.ac.il;

Tel.: +972 4 8294756; Fax: +972 4 8323041.

D. Burshtein and E. Weinstein: *Deptartment of Electrical Engineering - Systems, Tel-Aviv University, Tel-Aviv, Israel.*

Abstract

In this paper we present a theoretical analysis of the newly suggested TF–GSC algorithm. In recent work we considered a microphone array located in a reverberated room, where general transfer functions (TFs) relate the source signal and the microphones. The array is used for enhancing a speech signal contaminated by interference. The new, frequency-domain algorithm enables dealing with a complicated TF in the same simple manner as Griffiths & Jim GSC algorithm deals with delay-only arrays. It was shown that it is sufficient to use the ratio between the different TFs rather than the TFs themselves in order to implement the suggested algorithm. An unbiased estimate of the TFs ratios was achieved by exploiting the non-stationarity of the speech signal. Experimental results show that the algorithm is working well in various real-life scenarios and is outperforming the classical GSC algorithm.

In this contribution we turn to analytical evaluation of the performance of the suggested algorithm. We first derive a general expression of the enhancer output. We use this expression to evaluate two figures of merit, i.e. noise reduction ability and the amount of imposed distortion. The expression depends on the TFs involved, the noise field and the quality of estimation of the TFs ratios. For the latter dependency, we provide an approximated analysis of the signal's non-stationarity based estimation procedure and explore its dependency on the actual speech signal and the SNR level. It is shown, that for SNR levels as low as -5dB the estimation is almost bias free and imposes only minor distortion. This analysis is with good correspondence with previously achieved experimental results.

Keywords

Speech Enhancement, Beamforming, Non-Stationarity

I. Introduction

LATLY , an extension to the classical GSC [1], which deals with arbitrary transfer functions (TFs), was suggested [2],[3]. The time delay GSC (D–GSC) is widely analyzed in the literature [4],[5],[6] (for the two channel case), [7],[8],[9],[10] and [11]. Usually, a closed-form Wiener filter is used rather than the adaptive filters. The performance penalty for constraining the Wiener filters to an FIR structure is demonstrated by Nordholm *et al.* in [12]. The resulting performance limits depend on the cross-correlation between the sensors' signals induced by the noise field, as shown in the above references and by Cox [13].

Most references do not address the more practical scenario where complicated TFs relate the source and the sensors.

In this paper, we address the more general scenario by analyzing the TF–GSC structure [2]. For calculating the performance limits we use the closed form frequency domain solution (rather than the practical adaptive solution) for analyzing the suggested algorithm [14],[15]. This solution upper-bounds the expected performance of the adaptive solution. We examine the influence of the noise field and the TFs on the distortion imposed and the expected noise reduction.

We begin by briefly reviewing the TF–GSC in Section II. In Section III we formulate a general expression for the algorithm's output power spectral density (PSD). The desired signal *distortion* is addressed in Section IV, and the achievable *Noise reduction* and its dependency on the noise field is addressed in Section V. Evaluation of the suggested algorithm for special cases of noise fields and TFs will be presented in Section VI (distortion) and Section VII (noise reduction).

II. Summary of the TF–GSC

A. Problem Formulation

Consider an array of sensors in a noisy and reverberant environment. The received signal is comprised of two components. The first is some non-stationary (e.g. speech) signal. The second is some stationary interference signal. Our goal is to reconstruct the non-stationary signal component from the received signals. We use the following notation. $z_m(t)$ is the m-th sensor signal. $s(t)$ is the desired signal source. $n_m(t)$ is the interference signal of the m-th sensor. $n_m(t)$ is comprised of some coherent noise component and some ambient noise component. $a_m(t)$ are time-varying TFs from the desired signal source to the m-th sensor. The received signals are

given by,

$$z_m(t) = a_m(t) * s(t) + n_m(t)\ ;\ m = 1, \ldots, M \tag{1}$$

where $*$ denotes convolution. Using short term frequency analysis and assuming time-invariant TFs we have in the frequency domain in vector form,

$$\boldsymbol{Z}(t, e^{j\omega}) = \boldsymbol{A}(e^{j\omega}) S(t, e^{j\omega}) + \boldsymbol{N}(t, e^{j\omega}) \tag{2}$$

where

$$\begin{aligned}
\boldsymbol{Z}^T(t, e^{j\omega}) &= \left[\ Z_1(t, e^{j\omega}) \quad Z_2(t, e^{j\omega}) \quad \cdots \quad Z_M(t, e^{j\omega})\ \right] \\
\boldsymbol{A}^T(e^{j\omega}) &= \left[\ A_1(e^{j\omega}) \quad A_2(e^{j\omega}) \quad \cdots \quad A_M(e^{j\omega})\ \right] \\
\boldsymbol{N}^T(t, e^{j\omega}) &= \left[\ N_1(t, e^{j\omega}) \quad N_2(t, e^{j\omega}) \quad \cdots \quad N_M(t, e^{j\omega})\ \right]
\end{aligned}$$

and $Z_m(t, e^{j\omega}), S(t, e^{j\omega})$ and $N_m(t, e^{j\omega})$ are the short time Fourier transforms (STFT) of the respective signals. $A_m(e^{j\omega})$ is the frequency response of the m-th sensor TF, assumed to be time invariant during the analysis period.

B. Previously suggested algorithm

In [2] we introduced an approach for signal enhancement based on the desired signal non-stationarity. The M microphone signals are filtered by a corresponding set of M filters, $W^*(t, e^{j\omega})\ ; m = 1, \ldots, M$, and their outputs are summed to form the beamformer output:

$$Y(t, e^{j\omega}) = \boldsymbol{W}^\dagger(t, e^{j\omega}) \boldsymbol{Z}(t, e^{j\omega}),$$

where,

$$\boldsymbol{W}^\dagger(t, e^{j\omega}) = \left[\ W_1^*(t, e^{j\omega}) \quad W_2^*(t, e^{j\omega}) \quad \cdots \quad W_M^*(t, e^{j\omega})\ \right],$$

* denotes conjugation and † denotes conjugation transpose. $\boldsymbol{W}(t, e^{j\omega})$ is determined by minimizing the output power subject to the constraint that the signal portion of the output is the desired signal, $S(t, e^{j\omega})$, up to some pre-specified filter $\mathcal{F}^*(t, e^{j\omega})$ (usually a simple delay):

$$\min_{\boldsymbol{W}} \left\{ \boldsymbol{W}^\dagger(t, e^{j\omega}) \Phi_{\boldsymbol{ZZ}}(t, e^{j\omega}) \boldsymbol{W}(t, e^{j\omega}) \right\} \ \text{subject to}\ \ \boldsymbol{W}^\dagger(t, e^{j\omega}) \boldsymbol{A}(e^{j\omega}) = \mathcal{F}^*(t, e^{j\omega}). \tag{3}$$

This minimization can be efficiently implemented by constructing a *Generalized Sidelobe Canceller* (GSC) structure. By splitting

$$\boldsymbol{W}(t, e^{j\omega}) = \boldsymbol{W} - \mathcal{H}(e^{j\omega}) \boldsymbol{G}(t, e^{j\omega})$$

we decouple the role of minimization [by adjusting $\boldsymbol{G}(t, e^{j\omega})$] and constraining [by applying $\boldsymbol{W}_0(t, e^{j\omega})$] operations. Figure 1 summarizes the resulted algorithm. The output of the algorithm

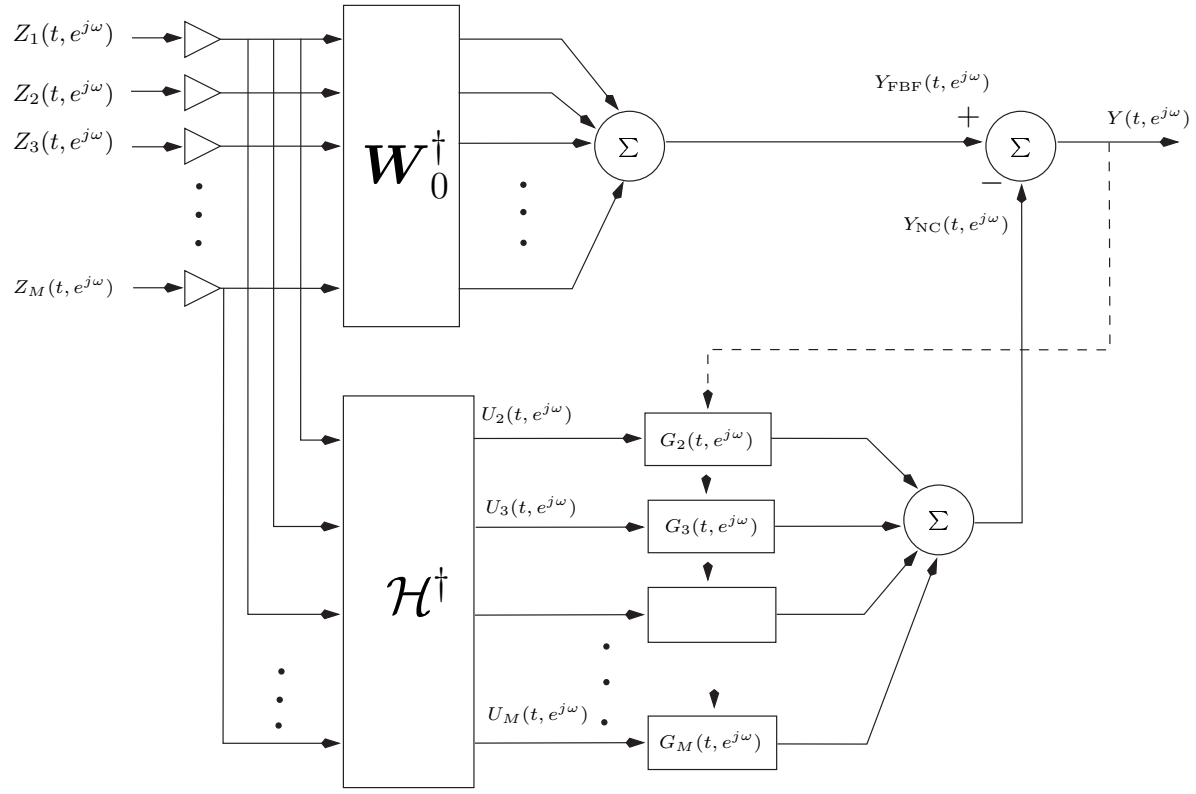


Fig. 1. GSC solution for the general TFs case (TF–GSC).

is given by,

$$Y(t, e^{j\omega}) = Y_{\mathrm{FBF}}(t, e^{j\omega}) - Y_{\mathrm{NC}}(t, e^{j\omega}) \tag{4}$$

where

$$\begin{aligned} Y_{\mathrm{FBF}}(t, e^{j\omega}) &= \boldsymbol{W}_0^{\dagger}(t, e^{j\omega})\boldsymbol{Z}(t, e^{j\omega}) \\ Y_{\mathrm{NC}}(t, e^{j\omega}) &= \boldsymbol{G}^{\dagger}(t, e^{j\omega})\mathcal{H}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t, e^{j\omega}) = \boldsymbol{G}^{\dagger}(t, e^{j\omega})\boldsymbol{U}(t, e^{j\omega}) \end{aligned} \tag{5}$$

The GSC solution is comprised of three components: a fixed beamformer (FBF) implemented by $\boldsymbol{W}_0^{\dagger}(t, e^{j\omega})$, a blocking matrix (BM) implemented by $\mathcal{H}^{\dagger}(e^{j\omega})$ that constructs the noise reference signals and a multi-channel noise canceller (NC) implemented by the filters $\boldsymbol{G}(t, e^{j\omega})$.

C. TFs ratio based enhancement

Although an exact knowledge of the TFs $\boldsymbol{A}(e^{j\omega})$ would yield distortionless reconstruction of the desired signal, it has been shown that knowledge of the TFs ratio may be sufficient in practice. Let,

$$H_m(e^{j\omega}) = \frac{A_m(e^{j\omega})}{A_1(e^{j\omega})} \; ; m = 1, \ldots, M \tag{6}$$

and

$$\boldsymbol{H}^T(e^{j\omega}) = \left[\begin{array}{cccc} 1 & H_2(e^{j\omega}) & \cdots & H_M(e^{j\omega}) \end{array} \right] = \frac{\boldsymbol{A}^T(e^{j\omega})}{A_1(e^{j\omega})}$$

A sub-optimal FBF block, which aligns the desired signal components but does not eliminate the reverberation term $A_1(e^{j\omega})$ can be implemented by,

$$\boldsymbol{W}_0(t, e^{j\omega}) = \frac{\boldsymbol{H}(e^{j\omega})}{\|\boldsymbol{H}(e^{j\omega})\|^2} \mathcal{F}(e^{j\omega}). \tag{7}$$

The blocking matrix must fulfill the equation,

$$\mathcal{H}^{\dagger}(e^{j\omega}) \boldsymbol{A}(e^{j\omega}) = 0. \tag{8}$$

Note that

$$\begin{aligned} \boldsymbol{U}(t, e^{j\omega}) &= \mathcal{H}^{\dagger}(e^{j\omega}) \boldsymbol{Z}(t, e^{j\omega}) \\ &= \mathcal{H}^{\dagger}(e^{j\omega}) [\boldsymbol{A}(e^{j\omega}) S(t, e^{j\omega}) + \boldsymbol{N}(t, e^{j\omega})] = \mathcal{H}^{\dagger}(e^{j\omega}) \boldsymbol{N}(t, e^{j\omega}) \end{aligned} \tag{9}$$

The last transition is due to (8). $\boldsymbol{U}(t, e^{j\omega})$ are noise only terms denoted *reference noise* signals. Consider the following $M \times (M-1)$ matrix $\mathcal{H}(e^{j\omega})$,

$$\mathcal{H}(e^{j\omega}) = \left[\begin{array}{cccc} -\frac{A_2^*(e^{j\omega})}{A_1^*(e^{j\omega})} & -\frac{A_3^*(e^{j\omega})}{A_1^*(e^{j\omega})} & \cdots & -\frac{A_M^*(e^{j\omega})}{A_1^*(e^{j\omega})} \\ 1 & 0 & \ldots & 0 \\ 0 & 1 & \ldots & 0 \\ & \ldots & \ddots & \\ 0 & 0 & \ldots & 1 \end{array} \right] \tag{10}$$

It can be easily verified that this matrix satisfies (8) and is thus a proper blocking matrix, that may be used for generating the reference noise signals, $\boldsymbol{U}(t, e^{j\omega})$. Hence, we have,

$$U_m(e^{j\omega}) = Z_m(t, e^{j\omega}) - \frac{A_m(e^{j\omega})}{A_1(e^{j\omega})} Z_1(t, e^{j\omega}) \; ; m = 2, \ldots, M \tag{11}$$

Thus, the knowledge of the TFs ratios $H_m(e^{j\omega}) = \frac{A_m(e^{j\omega})}{A_1(e^{j\omega})}$ is sufficient to implement the sidelobe canceller. The filters $\boldsymbol{G}(t, e^{j\omega})$ are adjusted to minimize the power at the output, $Y(t, e^{j\omega})$, exactly as in Widrow's classical problem [16]. A closed-form solution is obtained by using the multi-channel Wiener filter. Recall (4), our goal is to set $\boldsymbol{G}(t, e^{j\omega})$ so as to minimize

$$\mathrm{E}\left\{||Y_{\mathrm{FBF}}(t, e^{j\omega}) - \boldsymbol{G}^{\dagger}(t, e^{j\omega})\boldsymbol{U}(t, e^{j\omega})||^2\right\}$$

Let

$$\begin{aligned}\Phi_{\boldsymbol{U}Y}(t, e^{j\omega}) &= E\{\boldsymbol{U}(t, e^{j\omega})Y^{*}_{\mathrm{FBF}}(t, e^{j\omega})\} \\ \Phi_{\boldsymbol{U}\boldsymbol{U}}(t, e^{j\omega}) &= E\{\boldsymbol{U}(t, e^{j\omega})\boldsymbol{U}^{\dagger}(t, e^{j\omega})\},\end{aligned}$$

then the multi-channel Wiener filter is given by [12],[5]

$$\boldsymbol{G}(t, e^{j\omega}) = \Phi^{-1}_{\boldsymbol{U}\boldsymbol{U}}(t, e^{j\omega})\Phi_{\boldsymbol{U}Y}(t, e^{j\omega}) \tag{12}$$

So far we assumed that the TFs ratio vector, $\boldsymbol{H}(e^{j\omega})$, is known. However, In practice $\boldsymbol{H}(e^{j\omega})$ is not known and should be estimated. Rearranging terms in (11) we have

$$Z_m(t, e^{j\omega}) = H_m(e^{j\omega})Z_1(t, e^{j\omega}) + U_m(t, e^{j\omega}) \tag{13}$$

We assume that the TFs ratios are slowly changing in time compared to the time variations of the desired signal. We further assume that the statistics of the noise signal is slowly changing compared to the statistics of the desired signal. Consider some analysis interval during which both the TFs and the noise signal are assumed to be stationary. We divide this analysis interval into frames, such that the desired signal may be considered stationary during each frame. In [17] it is proposed to obtain an unbiased estimate of $H_m(e^{j\omega})$ by applying least squares to the following set of over-determined equations

$$\begin{bmatrix} \hat{\Phi}^{(1)}_{z_m z_1}(e^{j\omega}) \\ \hat{\Phi}^{(2)}_{z_m z_1}(e^{j\omega}) \\ \vdots \\ \hat{\Phi}^{(K)}_{z_m z_1}(e^{j\omega}) \end{bmatrix} = \begin{bmatrix} \hat{\Phi}^{(1)}_{z_1 z_1}(e^{j\omega}) & 1 \\ \hat{\Phi}^{(2)}_{z_1 z_1}(e^{j\omega}) & 1 \\ \vdots & \\ \hat{\Phi}^{(K)}_{z_1 z_1}(e^{j\omega}) & 1 \end{bmatrix} \begin{bmatrix} H_m(e^{j\omega}) \\ \Phi_{u_m z_1}(e^{j\omega}) \end{bmatrix} + \begin{bmatrix} \varepsilon^{(1)}_m(e^{j\omega}) \\ \varepsilon^{(2)}_m(e^{j\omega}) \\ \vdots \\ \varepsilon^{(K)}_m(e^{j\omega}) \end{bmatrix} \tag{14}$$

(a separate set of equations is used for each $m = 2, \ldots, M$). K is the number of frames used. $\Phi^{(k)}_{z_i z_j}(e^{j\omega})$ is the cross-PSD between z_i and z_j during the $k-th$ frame ($k = 1, \ldots, K$). $\Phi_{u_m z_1}(e^{j\omega})$ is the cross-PSD between u_m and z_1. $\hat{\Phi}^{(k)}_{z_i z_j}(e^{j\omega})$ and $\hat{\Phi}^{(k)}_{u_m z_1}(e^{j\omega})$ are the corresponding estimates.

III. Output Signal Power Spectrum Density

Using (4),(5) and (7) the algorithm's output is given by,

$$\begin{aligned} Y(t,e^{j\omega}) &= Y_{\mathrm{FBF}}(t,e^{j\omega}) - \boldsymbol{G}^{\dagger}(t,e^{j\omega})\boldsymbol{U}(t,e^{j\omega}) \\ &= \frac{\mathcal{F}^{*}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega}) - \boldsymbol{G}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega}), \end{aligned}$$

where only estimates of the TFs ratio, $\widehat{\boldsymbol{H}}(e^{j\omega})$ [and $\widehat{\mathcal{H}}(e^{j\omega})$], rather than the exact values, are assumed to be known. Using this expression, the PSD of the output signal can be calculated,

$$\begin{aligned} &\Phi_{yy}(t,e^{j\omega}) = E\{Y(t,e^{j\omega})Y^{*}(t,e^{j\omega})\} = \\ &E\left\{\left(\frac{\mathcal{F}^{*}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega}) - \boldsymbol{G}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega})\right)\times\right. \\ &\left.\left(\frac{\mathcal{F}^{*}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega}) - \boldsymbol{G}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{Z}(t,e^{j\omega})\right)^{\dagger}\right\}. \end{aligned}$$

Opening brackets and using the PSD definition $\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega}) = E\{\boldsymbol{Z}(t,e^{j\omega})\boldsymbol{Z}^{\dagger}(t,e^{j\omega})\}$ yields,

$$\begin{aligned} \Phi_{yy}(t,e^{j\omega}) &= \frac{|\mathcal{F}(e^{j\omega})|^{2}}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{4}}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}) \qquad (15) \\ &- \frac{\mathcal{F}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\boldsymbol{G}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}) \\ &- \frac{\mathcal{F}^{*}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\boldsymbol{H}}^{\dagger}(t,e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\boldsymbol{G}(e^{j\omega}) \\ &+ \boldsymbol{G}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\boldsymbol{G}(e^{j\omega}). \qquad (16) \end{aligned}$$

The output PSD depends on the input signal $\boldsymbol{Z}(t,e^{j\omega})$ and the optimal multi-channel Wiener filter given by (12). Although signal leakage might occur due to errors in estimating $\boldsymbol{H}(e^{j\omega})$, we can calculate the filter during non-active periods of the desired signal, in which $\boldsymbol{Z}(t,e^{j\omega}) = \boldsymbol{N}(t,e^{j\omega})$. Thus,

$$\boldsymbol{U}(t,e^{j\omega}) = \widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{N}(t,e^{j\omega}).$$

Calculating the involved PSDs

$$\begin{aligned} \Phi_{\boldsymbol{U}Y}(t,e^{j\omega}) &= E\{\boldsymbol{U}(t,e^{j\omega})Y_{\mathrm{FBF}}^{*}(t,e^{j\omega})\} \\ &= E\left\{\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{N}(t,e^{j\omega})\left(\frac{\mathcal{F}^{*}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\boldsymbol{N}(t,e^{j\omega})\right)^{\dagger}\right\} \\ &= \frac{\mathcal{F}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^{2}}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}) \end{aligned}$$

and

$$\begin{aligned}\Phi_{\boldsymbol{UU}}(t,e^{j\omega}) &= E\{\boldsymbol{U}(t,e^{j\omega})\boldsymbol{U}^{\dagger}(t,e^{j\omega})\}\\ &= E\left\{\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\boldsymbol{N}(t,e^{j\omega})\boldsymbol{N}^{\dagger}(t,e^{j\omega})\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\right\}\\ &= \widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega}),\end{aligned}$$

where $\Phi_{\boldsymbol{NN}}(t,e^{j\omega})=E\{\boldsymbol{N}(t,e^{j\omega})\boldsymbol{N}^{\dagger}(t,e^{j\omega})\}$, we obtain,

$$\boldsymbol{G}(t,e^{j\omega})=\frac{\mathcal{F}(e^{j\omega})}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^2}\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}). \quad (17)$$

Using desired signal and noise signal independence, the *noise reduction* and the *distortion* imposed by the algorithm, can be calculated separately by deriving expressions for the output PSD in the following two situations:

$$\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})=\begin{cases}\Phi_{\boldsymbol{NN}}(t,e^{j\omega}) & \text{no signal}\\ \Phi_{ss}(t,e^{j\omega})\boldsymbol{A}(e^{j\omega})\boldsymbol{A}^{\dagger}(e^{j\omega}) & \text{no noise}\end{cases}$$

yielding,

$$\Phi_{yy}(t,e^{j\omega})=\begin{cases}\Phi_{yy}^{n}(t,e^{j\omega}) & \text{no signal} \;\Rightarrow\; \text{noise reduction}\\ \Phi_{yy}^{s}(t,e^{j\omega}) & \text{no noise} \;\Rightarrow\; \text{distortion}\end{cases}$$

Using (17) and (15) we obtain the output signal output PSD given in (18).

$$\begin{aligned}\Phi_{yy}(t,e^{j\omega}) &= \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^4}\times\Big\{\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega})-\\ &\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega})-\\ &\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega})+\\ &\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\\ &\times\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega})\Big\}\end{aligned} \quad (18)$$

This complicated expression depends on various parameters: the input signal PSD, $\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega})$, the noise PSD used for calculating the optimal filters, $\Phi_{\boldsymbol{NN}}(t,e^{j\omega})$, and the TFs' ratio estimate $\widehat{\boldsymbol{H}}(e^{j\omega})$ [which is also used for the blocking matrix $\widehat{\mathcal{H}}(e^{j\omega})$]. This expression will be used is Sections IV,V for deriving general expressions for the distortion imposed by the algorithm and noise reduction achieved, respectively. In Sections VI,VII these general expressions will be evaluated for several interesting cases.

IV. Desired Signal Distortion

The distortion imposed by the algorithm can be calculated by the general expression given in (18) for a signal $\boldsymbol{Z}(t,e^{j\omega}) = \boldsymbol{A}(e^{j\omega})s(t,e^{j\omega})$. Assume we have exact knowledge of the TFs' ratio $\boldsymbol{H}(e^{j\omega})$, i.e., $\widehat{\boldsymbol{H}}(e^{j\omega}) = \boldsymbol{H}(e^{j\omega}) = \frac{\boldsymbol{A}(e^{j\omega})}{A_1(e^{j\omega})}$. Thus, using the signal spectrum expression $\Phi_{\boldsymbol{ZZ}}(t,e^{j\omega}) = \Phi_{ss}(t,e^{j\omega})\boldsymbol{A}(e^{j\omega})\boldsymbol{A}^{\dagger}(e^{j\omega})$ and the identity $\mathcal{H}^{\dagger}(e^{j\omega})\boldsymbol{A}(e^{j\omega}) = 0$, expression (18) reduces to

$$\begin{aligned}\Phi^{s}_{yy}(t,e^{j\omega}) &= \Phi_{ss}(t,e^{j\omega})\frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\boldsymbol{H}^{\dagger}(e^{j\omega})\boldsymbol{A}(e^{j\omega})\boldsymbol{A}^{\dagger}(e^{j\omega})\boldsymbol{H}(e^{j\omega}) \\ &= \Phi_{ss}(t,e^{j\omega})|\mathcal{F}(e^{j\omega})|^2|A_1(e^{j\omega})|^2. \end{aligned} \tag{19}$$

The filter $\mathcal{F}(e^{j\omega})$ is an arbitrary predetermined filter, so it should not be regarded as a distortion. The filter $A_1(e^{j\omega})$ is the TF from the signal source to the first (reference) sensor. It can not be eliminated by the algorithm. Actually, it imposes on the output signal the same amount of distortion imposed on the arbitrary reference sensor. Thus, we define the total distortion of the algorithm by normalizing the output,

$$\mathrm{DIS}(t,e^{j\omega}) = \frac{\Phi^{s}_{yy}(t,e^{j\omega})}{|\mathcal{F}(e^{j\omega})|^2|A_1(e^{j\omega})|^2\Phi_{\boldsymbol{ss}}(t,e^{j\omega})}.$$

A value of $\mathrm{DIS}(t,e^{j\omega}) = 1$ indicates a distortionless output. This value is achieved whenever an exact knowledge of the TFs ratios is available. However, in actual scenarios the filters $\boldsymbol{H}(e^{j\omega})$ are not known in advance, i.e.

$$\widehat{\boldsymbol{H}}(e^{j\omega}) = \boldsymbol{H}(e^{j\omega}) + \boldsymbol{E}(e^{j\omega}) \neq \boldsymbol{H}(e^{j\omega})$$

In this case, (18) is used again, this time with the estimated TFs ratios, to calculate the amount of distortion. This estimation error has twofold influence. First, the FBF is not accurate, so it could degrade the alignment of the signal, causing incoherent addition. Second, the blocking matrix, whose terms depend on $\boldsymbol{H}(e^{j\omega})$ estimate, would not block the desired signal completely, causing self-cancellation. The error depends, of course, on the estimation method used. Analysis of the estimation error, $\boldsymbol{E}(e^{j\omega})$, is given in [17]. The Blackman-Tukey approach for PSD estimation, involved in the procedure, is used. PSD estimation is obtained by multiplying the empirical correlation function (in each frame) by a window function. Frame errors are assumed to be mutually uncorrelated. The resulting TFs ratio estimates, using the nonstationarity equations,

are bias free, and the variance of the estimation is given by [17],

$$\begin{aligned}\mathrm{v}ar\{E^m(e^{j\omega})\} &= \frac{\Phi_{u_mu_m}(t,e^{j\omega})}{BT}\frac{<1/\Phi_{z_1z_1}(t,e^{j\omega})>}{<\Phi_{z_1z_1}(t,e^{j\omega})><1/\Phi_{z_1z_1}(t,e^{j\omega})>-1} \\ &\quad \frac{1}{BT}\frac{\Phi_{u_mu_m}(t,e^{j\omega})}{<\Phi_{z_1z_1}(t,e^{j\omega})>}\frac{<\Phi_{z_1z_1}(t,e^{j\omega})><1/\Phi_{z_1z_1}(t,e^{j\omega})>}{<\Phi_{z_1z_1}(t,e^{j\omega})><1/\Phi_{z_1z_1}(t,e^{j\omega})>-1}\end{aligned}$$

where for a given set of K values $\beta^{(k)}(e^{j\omega})$ the average operator is defined by,

$$\left\langle \beta(e^{j\omega}) \right\rangle \stackrel{\triangle}{=} \frac{1}{K}\sum_{k=1}^{K}\beta^{(k)}(e^{j\omega}).$$

T is the total time used for estimation, $B = \frac{1}{\sum_\tau w^2(\tau)}$ is related to the window bandwidth. Note, that a conventional estimation procedure, $\widehat{\boldsymbol{H}}(e^{j\omega}) = \frac{\Phi_{z_mz_1}(t,e^{j\omega})}{\Phi_{z_1z_1}(t,e^{j\omega})}$ is biased in our problem and its variance is given by the well-known expression (see, for instance, [17]),

$$\mathrm{v}ar\{E^m_{Conv}(e^{j\omega})\} = \frac{1}{BT}\frac{\Phi_{u_mu_m}}{<\Phi_{z_1z_1}(t,e^{j\omega})>}.$$

Define *relative efficiency* (the ratio between nonstationarity method and conventional method variances),

$$\Xi(e^{j\omega}) = \frac{<\Phi_{z_1z_1}(t,e^{j\omega})><1/\Phi_{z_1z_1}(t,e^{j\omega})>}{<\Phi_{z_1z_1}(t,e^{j\omega})><1/\Phi_{z_1z_1}(t,e^{j\omega})>-1}.$$

Note, that as $\Phi_{z_1z_1}(t,e^{j\omega}) \geq 0$, the relative efficiency always fulfills $\Xi(e^{j\omega}) > 1$. For a stationary signal $\Xi(e^{j\omega}) \to \infty$. Thus, $1 < \Xi(e^{j\omega}) < \infty$. The more the signal is nonstationary, $\Xi(e^{j\omega})$ tends to 1, and the more it is stationary $\Xi(e^{j\omega})$ tends to infinity. Define, averaged signal to noise ratio,

$$SNR^m_{ave}(e^{j\omega}) = \frac{<\Phi_{z_1z_1}(t,e^{j\omega})>}{\Phi_{u_mu_m}(t,e^{j\omega})}$$

As,

$$Z_m(t,e^{j\omega}) = H_m(e^{j\omega})Z_1(t,e^{j\omega}) + U_m(t,e^{j\omega})$$

the averaged SNR depends on the power ratio between the input signal, $z_1(t)$ and the related disturbance, $u_m(t)$. Collecting all definitions we obtain,

$$\mathrm{v}ar\{E^m(e^{j\omega})\} = \frac{1}{BT}\frac{1}{SNR^m_{ave}(e^{j\omega})}\Xi(e^{j\omega}). \tag{20}$$

The estimation error variance is increasing as the input signal $[z_1(t)]$ becomes more stationary, and as the averaged SNR decreases. Note, that if the first sensor noise signal, $n_1(t)$, is very strong, both $z_1(t)$ and $u_m(t)$ become large and the averaged SNR tends to 1. The degradation in the TFs' estimate is then due to the term $\Xi(e^{j\omega})$. When the (presumably stationary) noise

signal becomes dominant, the signal $z_1(t)$ becomes more stationary, causing $\Xi(e^{j\omega}) \to \infty$. This phenomenon may be interpreted in the following manner. As the signals approach stationarity, all the equations in the over-determined set given in (14) become similar, yielding an ill-conditioned problem.

Evaluation of the desired signal distortion, caused by the TFs ratios estimation error, for various scenarios is given in Section VI.

V. Noise Reduction

To calculate the noise reduction of the algorithm, we will use again the general expression for the output signal given by (18), this time with a noise signal $\boldsymbol{Z}(t, e^{j\omega}) = \boldsymbol{N}(t, e^{j\omega})$ (the same noise signal used to calculate the optimal Wiener filter) as the input signal. The expression (18) reduces to

$$\Phi^n_{yy}(t, e^{j\omega}) = \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^4}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}) - \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^4}\times \tag{21}$$
$$\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\left(\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\widehat{\mathcal{H}}(e^{j\omega})\right)^{-1}\widehat{\mathcal{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}).$$

Identify the first term of the equation as $\Phi^n_{fbf}(t, e^{j\omega})$, the FBF part of the output PSD,

$$\begin{aligned}\Phi^n_{fbf}(t, e^{j\omega}) &= E\{Y^n_{\mathrm{FBF}}(t, e^{j\omega})\left(Y^n_{\mathrm{FBF}}(t, e^{j\omega})\right)^*\} \\ &= \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\widehat{\boldsymbol{H}}(e^{j\omega})\|^4}\widehat{\boldsymbol{H}}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\widehat{\boldsymbol{H}}(e^{j\omega}).\end{aligned} \tag{22}$$

Another interesting figure of merit is the extra noise reduction achieved by the noise canceling branch (see also [7]),

$$\mathrm{NR}_{\mathrm{nc}}(t, e^{j\omega}) = \frac{\Phi^n_{fbf}(t, e^{j\omega})}{\Phi^n_{yy}(t, e^{j\omega})}. \tag{23}$$

Expressions (21), (22) and (23) can be used for calculating the noise reduction achieved by the algorithm and to determine the major contributor for this noise reduction. As usually the error in estimating $\boldsymbol{H}(e^{j\omega})$ is quite small and has minor influence on the noise reduction, we will assume throughout the noise reduction analysis perfect knowledge of the TFs ratios, i.e. $\widehat{\boldsymbol{H}}(e^{j\omega}) = \boldsymbol{H}(e^{j\omega})$.

A. Dependence on Noise Field

The resulting expression for the noise cancellation depends on the noise PSD at the sensors. We calculate the expected noise reduction of the algorithm for three important noise fields:

coherent (point source), diffused (spatially extended) and incoherent (noise signals generated at the sensors, e.g. amplifier noise, are assumed to be uncorrelated).

A.1 Coherent Noise Field

Assume a single point source noise signal with PSD $\Phi_{nn}(t, e^{j\omega})$ and assume that $b_m(t)$ are slowly time varying transfer functions relating the noise source and the m-th sensor. Define,

$$\boldsymbol{N}(t, e^{j\omega}) = \boldsymbol{B}(e^{j\omega})N(t, e^{j\omega})$$

where,

$$\boldsymbol{B}^T(e^{j\omega}) = \left[\begin{array}{cccc} B_1(e^{j\omega}) & B_2(e^{j\omega}) & \cdots & B_M(e^{j\omega}) \end{array} \right].$$

The PSD matrix of the noise component of the sensors signals is given by,

$$\Phi_{\boldsymbol{NN}}(t, e^{j\omega}) = \Phi_{nn}(t, e^{j\omega})\boldsymbol{B}(e^{j\omega})\boldsymbol{B}^{\dagger}(e^{j\omega}) + \varepsilon I$$

where I is an $M \times M$ identity matrix, and $\varepsilon \to 0$. The last term is added for stability reasons (see Appendix A). For $\boldsymbol{B}(e^{j\omega}) \neq \boldsymbol{A}(e^{j\omega})$, the achievable noise reduction is infinite, i.e.

$$\Phi^n_{yy}(t, e^{j\omega}) = 0 \ \text{ for } \ \boldsymbol{B}(e^{j\omega}) \neq \boldsymbol{A}(e^{j\omega}).$$

Thus, perfect noise cancellation is achieved. The derivation of this result is given in Appendix A. Note, that this is not a surprising result, since for $M \geq 2$ the Wiener filter can completely eliminate the noise component. This result is valid for all TFs $\boldsymbol{B}(e^{j\omega})$ except for the signal direction TFs $\boldsymbol{B}(e^{j\omega}) = \boldsymbol{A}(e^{j\omega})$. If $\boldsymbol{B}(e^{j\omega}) = \boldsymbol{A}(e^{j\omega})$ the noise and desired signal are indistinguishable and no noise reduction is expected. If $\boldsymbol{B}(e^{j\omega}) \neq \boldsymbol{A}(e^{j\omega})$ the suggested algorithm can eliminate any point source noise signal as good as the simple GSC can eliminate a directional noise signal in the delay-only propagation case (see for instance [7]). It is also interesting to evaluate the noise part of the FBF branch,

$$\begin{aligned} \Phi^n_{fbf}(t, e^{j\omega}) &= \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\boldsymbol{H}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\boldsymbol{H}(e^{j\omega}) \\ &= \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\boldsymbol{H}^{\dagger}(e^{j\omega})\Phi_{nn}(t, e^{j\omega})\boldsymbol{B}(e^{j\omega})\boldsymbol{B}^{\dagger}(e^{j\omega})\boldsymbol{H}(e^{j\omega}) \\ &= \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\Phi_{nn}(t, e^{j\omega})\boldsymbol{H}^{\dagger}(e^{j\omega})\boldsymbol{B}(e^{j\omega})\left(\boldsymbol{H}^{\dagger}(e^{j\omega})\boldsymbol{B}(e^{j\omega})\right)^{\dagger}. \end{aligned}$$

Although the FBF combines the desired signal coherently, it does not necessarily improves the SNR. The infinite noise reduction is due to the noise canceller branch.

A.2 Diffused Noise Field

In highly reverberant acoustical environment, such as a car enclosure, the noise field tends to be diffused (see for instance [18] and [5]). A diffused noise source is assumed to be equidistributed on a sphere in the far field of the array. The cross-coherence function between signals received by two sensors (i,j) with distance d_{ij} can be found in [18], and is given in the following expression:

$$\Gamma_{N_iN_j}(e^{j\omega}) = \frac{\Phi_{N_iN_j}(e^{j\omega})}{\sqrt{\Phi_{N_iN_i}(e^{j\omega})\Phi_{N_jN_j}(e^{j\omega})}} = \frac{\sin(\omega d_{ij}/c)}{\omega d_{ij}/c},$$

where c is the speed of sound. Thus, the coherence matrix is given by,

$$\Gamma(e^{j\omega}) = \begin{bmatrix} 1 & \Gamma_{N_1N_2}(e^{j\omega}) & \cdots & \Gamma_{N_1N_M}(e^{j\omega}) \\ \Gamma_{N_2N_1}(e^{j\omega}) & 1 & \cdots & \\ & & \ddots & \\ \Gamma_{N_MN_1}(e^{j\omega}) & & & 1 \end{bmatrix}.$$

The noise spectrum at the sensors input is thus,

$$\Phi_{\boldsymbol{NN}}(t, e^{j\omega}) = \Phi_{nn}(t, e^{j\omega})\Gamma(e^{j\omega}).$$

Using (22) the noise spectrum at the FBF output is given by,

$$\Phi^n_{fbf}(t, e^{j\omega}) = \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\Phi_{nn}(t, e^{j\omega})\boldsymbol{H}^\dagger(e^{j\omega})\Gamma(e^{j\omega})\boldsymbol{H}(e^{j\omega}).$$

The extra noise reduction achieved by the noise canceller is given by (24).

$$\mathrm{NR}_{\mathrm{nc}}(t, e^{j\omega}) = \left(1 - \frac{\boldsymbol{H}^\dagger(e^{j\omega})\Gamma(e^{j\omega})\mathcal{H}(e^{j\omega})\left(\mathcal{H}^\dagger(e^{j\omega})\Gamma(e^{j\omega})\mathcal{H}(e^{j\omega})\right)^{-1}\mathcal{H}^\dagger(e^{j\omega})\Gamma(e^{j\omega})\boldsymbol{H}(e^{j\omega})}{\boldsymbol{H}^\dagger(e^{j\omega})\Gamma(e^{j\omega})\boldsymbol{H}(e^{j\omega})}\right)^{-1}. \quad (24)$$

This expression depends on the TFs ratios $\boldsymbol{H}(e^{j\omega})$, assumed to be error free, and the coherence function $\Gamma(e^{j\omega})$.

A.3 Incoherent Noise Field

For incoherent noise field we assume that the noise at the sensors has no spatial correlation.

$$\Phi_{\boldsymbol{NN}}(t, e^{j\omega}) = \Phi_{nn}(t, e^{j\omega})I,$$

where I is an $M \times M$ identity matrix. Using (21) with perfect knowledge of the TFs ratio, i.e. $\widehat{\boldsymbol{H}}(e^{j\omega}) = \boldsymbol{H}(e^{j\omega})$, and with the prespecified $\Phi_{\boldsymbol{NN}}(t, e^{j\omega})$ we obtain,

$$\Phi^n_{yy}(t, e^{j\omega}) = \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\Phi_{nn}(t, e^{j\omega})\boldsymbol{H}^\dagger(e^{j\omega})\left\{I - \mathcal{H}\left(e^{j\omega})(\mathcal{H}^\dagger(e^{j\omega})\mathcal{H}(e^{j\omega})\right)^{-1}\mathcal{H}^\dagger(e^{j\omega})\right\}\boldsymbol{H}(e^{j\omega}).$$

It can be easily verified that,

$$\boldsymbol{H}^\dagger(e^{j\omega})\mathcal{H}(e^{j\omega}) = \mathbf{0}_{1\times(M-1)}.$$

Furthermore, as $\mathcal{H}^\dagger(e^{j\omega})\mathcal{H}(e^{j\omega})$ is a positive matrix, its inverse always exists. Thus, the contribution of the noise canceling branch is zero, and the noise reduction is only due to the fixed beamformer. The noise power at the output is thus,

$$\Phi^n_{yy}(t, e^{j\omega}) = \Phi^n_{fbf}(t, e^{j\omega}) = \Phi_{nn}(t, e^{j\omega})\frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4}\boldsymbol{H}^\dagger(e^{j\omega})\boldsymbol{H}(e^{j\omega}) = \Phi_{nn}(t, e^{j\omega})\frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^2}.$$

Again, no noise reduction is guaranteed by this structure, and the result depends on the TFs' ratio involved.

Evaluation of the noise reduction achieved by the algorithm is given in Section VII.

VI. Performance Evaluation: Distortion

In Section IV we observed that knowledge of the exact TFs yields a distortionless output. In this section, the influence of errors in estimating the TFs will be determined. The expected distortion imposed by the algorithm is evaluated. The results depend on the desired signal's TFs and the noise field.

We begin with the simple steering array, where only delays relate the sources and the sensors. Thus, the *direction of arrival* (DoA) of the sources completely determines the TFs. We determine the performance degradation as a function of the steering error. Then, dealing with the more general TFs case, we will use the estimation error variance (20) to predict the amount of distortion.

A. Signal's TFs: Pure Delay

Assume free space propagation, i.e., only pure delay relates the desired signal source and the sensors. Assume that the desired signal impinges the array from $\theta = 90^o$ and that the inter-element distances as depicted in Figure 3. The distortion imposed by the algorithm is evaluated by steering the array around the desired signal DoA (i.e. changing $\widehat{\boldsymbol{H}}(e^{j\omega})$ in the range $\theta \in [85^o, 95^o]$).

A.1 Directional Noise Signal

In the directional noise field case we optimize the array to cancel a noise source from $\theta = 40^o$ (by optimization of the array we refer to designing the optimal Wiener filter in the noise cancellation branch). In Figure 2 (top-left) we present the PSD of the output signal of the array as a function of the frequency and the steering angle. It is clearly shown that the output signal is distorted as we move the steering angle of the array away from the desired signal direction $\theta = 90^o$. As shown,

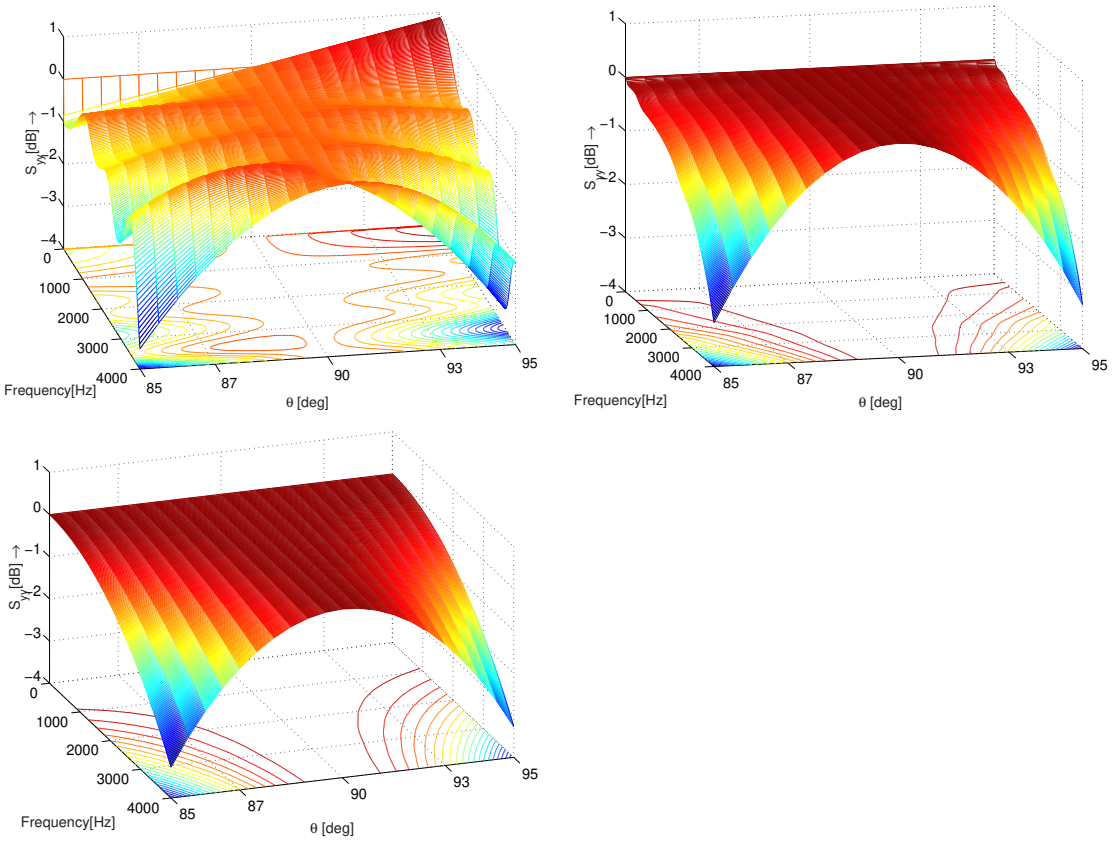


Fig. 2. Distortion as a function of the frequency and direction of arrival. Desired signal direction $\theta = 90^o$. $M = 5$ sensors. Top-left: Directional noise field ($\theta = 40^o$). Top-right: Diffused noise field. Bottom: Incoherent noise field.

the imposed distortion is not exceeding 4dB. Note that distortion can also cause an increase in the output PSD, since unwanted components leak into the algorithm's output.

A.2 Diffused Noise Field

If the array is designed to work with a diffused noise field, the distortion imposed on the desired signal is given in Figure 2 (top-right). Again, no more than 4dB distortion is imposed in

the rang $\theta \in [85, 95]$. Note the symmetric behavior of the output spectrum as a function of the steering error. This is due to the inherent symmetry in the noise field.

A.3 Incoherent Noise field

The same trend can be observed for an incoherent noise field, as seen in Figure 2 (bottom-left).

B. Signal's TFs: Arbitrary

Although we have a closed form formula for estimating the TFs' ratio, calculating the predicted distortion of the algorithm for the case of an arbitrary TFs is a complex task, since the performance depends on the actual TFs used. For this purpose we use the acoustical TFs (ATFs), which were obtained from real room recordings. The scenario shown in Figure 3 was studied. The enclosure is a conference room with dimensions $5m \times 4m \times 2.8m$. A linear array was placed

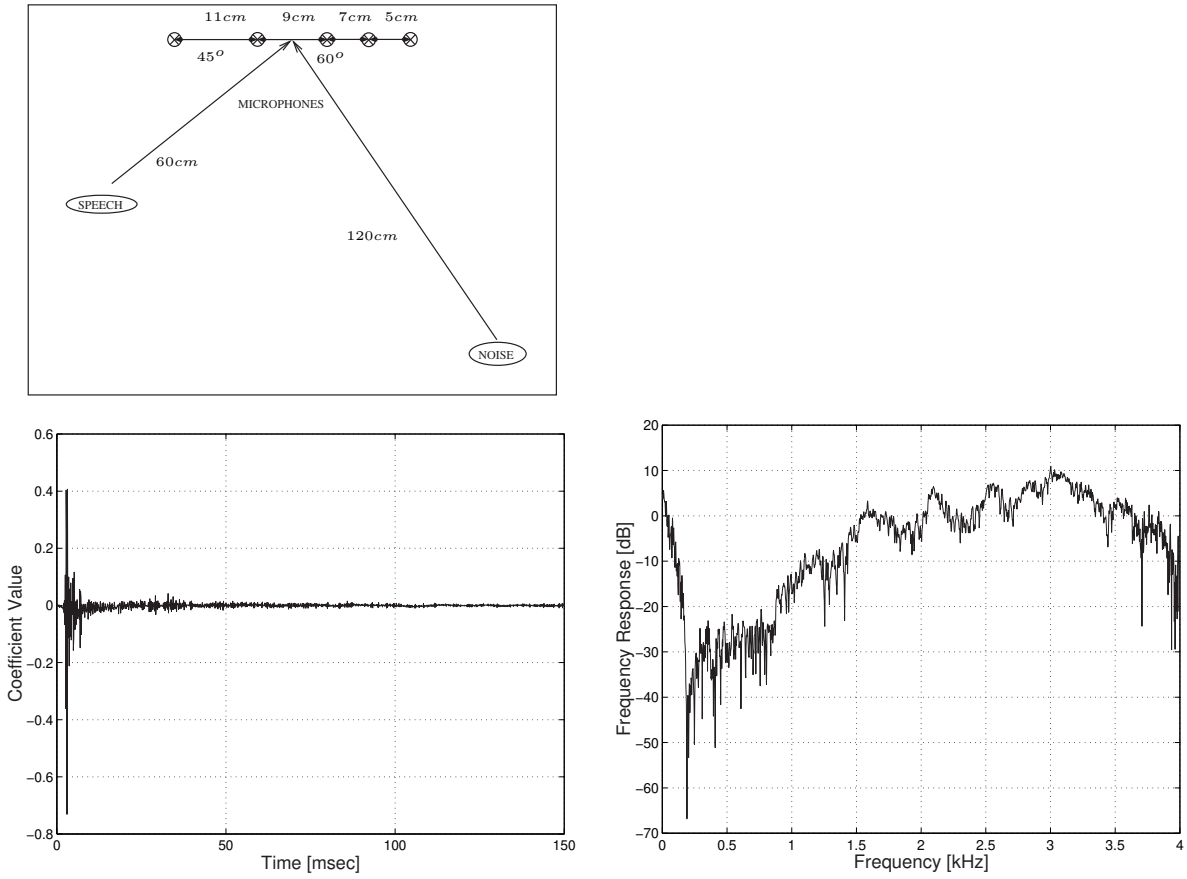


Fig. 3. Five microphones array in a noisy conference room. Test scenario (top), impulse response (bottom-left) and frequency response (bottom-right) of the ATF between the speech loudspeaker and the first microphone.

on a table at the center of the room. Two loudspeakers were used. One for the speech source and the other for the noise source. The sources and microphones locations, the impulse response and frequency response between the speech source and the first microphone are depicted in Figure 3. This response was obtained using a least squares fit between the input signal source and the received microphone signal (the response includes the loudspeaker). We note that in all our experiments we used the actual recordings and did not use the estimated impulse responses.

The estimation error also depends on the estimation procedure used. We evaluate the suggested estimation procedure, based on signal nonstationarity. The estimated TFs are bias-free, but with increased variance in comparison with the standard LS method. The error variance is given by (20). Note, that in order to calculate the error variance we need the reference noise signals $u_m(t);\ m = 1, \ldots, M$. These signals themselves depend on the blocking matrix, which terms depend on the estimated TFs. Thus, the problem of predicting the distortion becomes difficult to solve. Furthermore, as each TF estimation depends on the signal $z_1(t)$, a correlation between the estimation errors of each filter results.

Several simplifications and assumptions enable a rough estimation of the expected distortion:

- The errors in estimating each TF are uncorrelated.
- The signals $u_m(t)$ are only slightly changed due errors in estimating the TFs, so reestimating the predicted error with the new noise signals is unnecessary. This assumption states that we will use the exact TFs $H(e^{j\omega})$ for calculating the reference signals $u_m(t)$ PSD, and use the resulting PSDs to calculate the imposed error in estimating $H(e^{j\omega})$. This assumption may be referred to as *small error assumption*.
- The estimation error is Gaussian distributed with zero mean (unbiased estimate) and its variance is given by (20).

The estimation error PSD depends on the signal nonstationarity. A typical nonstationary function, $\Xi(e^{j\omega})$ for clean and noisy speech signals is given in Figure 4. It is clearly shown, that except for outliers in several frequencies, the clean signal is quite nonstationary (values range between 1 and 1.5). The noisy signal becomes more stationary, yielding an increased value of $\Xi(e^{j\omega})$, as shown in Figure 4. Although the error variance is likewise dependent on the noise field, our experiments show only weak dependence on it. Thus, we give only the results for noise field caused by a point source propagating through ATFs. The desired signal uses one set of ATFs (originating from left side of the array) and the noise signal uses another set

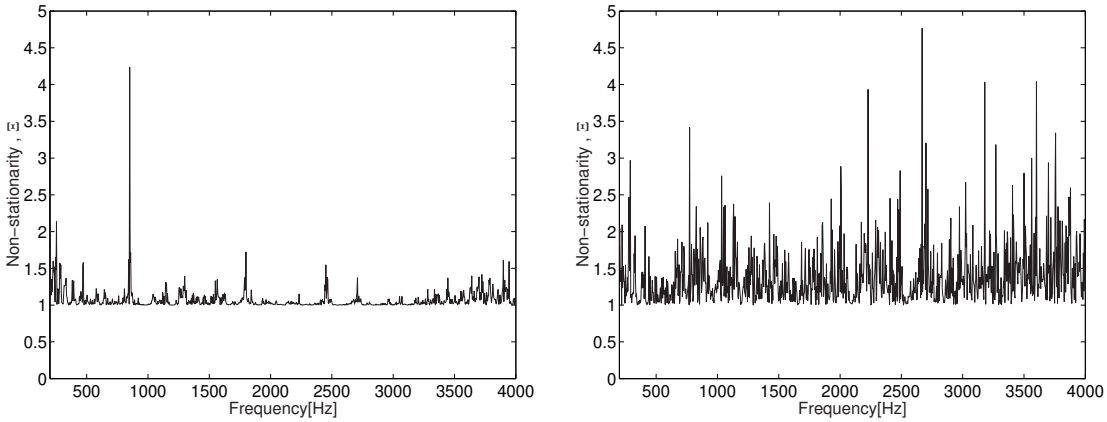


Fig. 4. Typical nonstationarity function for speech signal. Clean (left). Noisy SNR=-5 dB (right).

(originating from right side of the array). Distortion plots are shown in Figure 5 for various SNR levels. The plots were obtained by averaging over 50 *Monte-Carlo* experiments. It is evident that the amount of distortion imposed is quite low and even for low input SNR of -5dB, the predicted distortion is no more than 4dB in the interesting frequency band. Although we made some restrictive assumptions, the low distortion predicted by this analysis may be noticed while practically applying our algorithm to a speech signal, as shown in [2].

VII. Performance Evaluation: Noise Reduction

To evaluate the expected noise reduction of the algorithm, as we evaluated the predicted distortion, we should determine the TFs involved and the noise field. We begin with the simple steering array, where only delays relate the sources and the sensors. Thus, the DoA of the sources determines completely the TFs. Then, we deal with the more general TFs case. Throughout the noise reduction evaluation we assume that the TFs' ratios are exactly known.

A. Signal's TFs: Pure Delay

Assume free space propagation, i.e., only pure delay relates the desired signal source and the sensors. The inter-element distances are shown in Figure 3. Signal is assumed to impinge the array from $\theta = 90^o$

A.1 Directional Noise Signal

We optimize the array to cancel noise source from $\theta = 40^o$. In Figure 6 (left side) we present the output signal of the array as a function of the frequency and the array steering angle θ. It

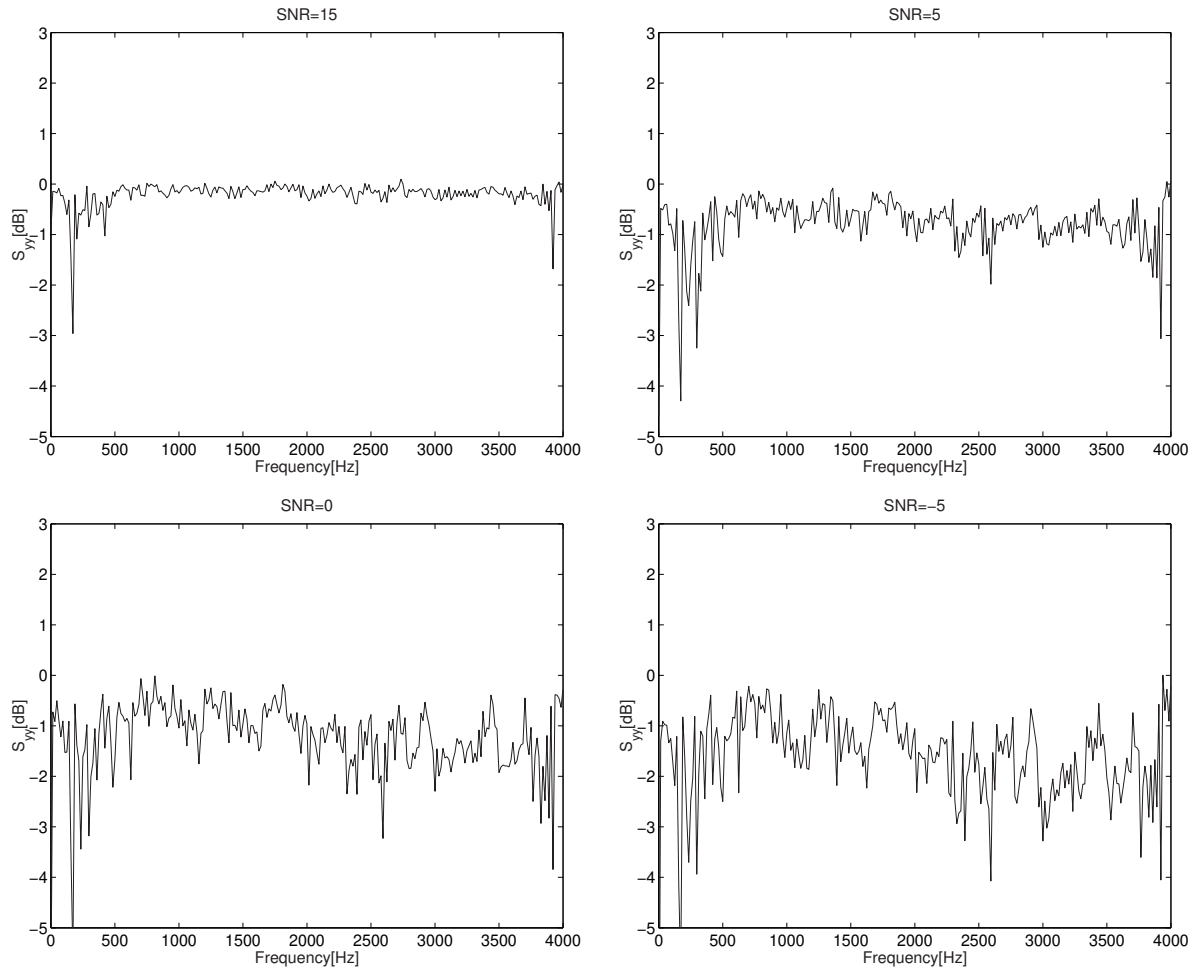


Fig. 5. Distortion for arbitrary ATF.

is clearly shown that the main lobe is maintained, while null is constructed in all frequencies at the noise angle. The main lobe is wider in the lower frequency band. This is in good agreement with the theory, since at $\omega = 0$ [rad/sec] there is no phase difference between the signals at the sensors.

A.2 Diffused Noise Field

The array configuration depicted in Figure 3 is analyzed with diffused noise field by using (24). In Figure 7 we evaluate the extra noise reduction of the noise canceller branch for four different steering angles as a function of the frequency and the number of sensors. Compare the results with [5] and [4]. The case of $M = 5$ sensors is shown in Figure 6 (right side) for various steering angles and the entire frequency band. It is clear that almost no noise reduction is obtained in

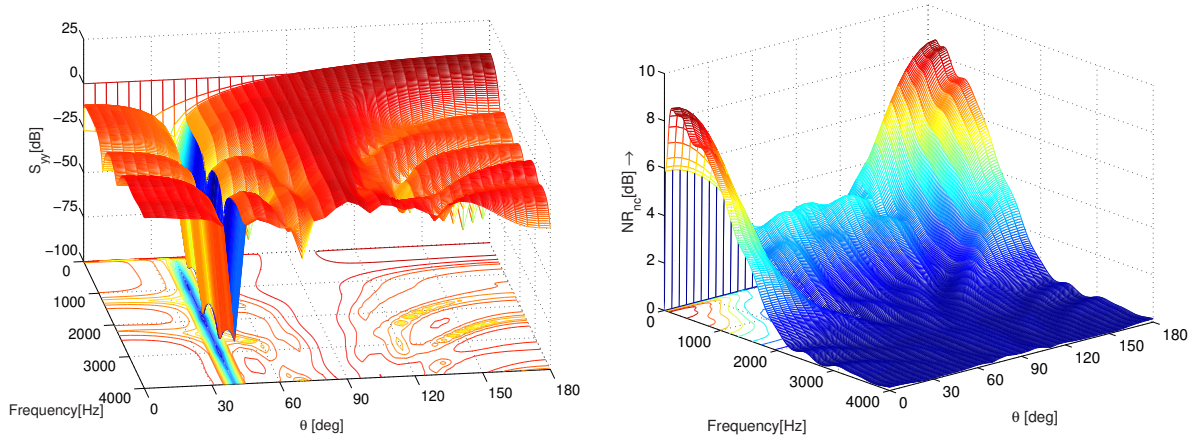


Fig. 6. Linear array with $M = 5$ sensors for delay-only ATFs. Left: Array output for directional noise field. Right: Extra noise reduction of noise cancelling branch for diffused noise field.

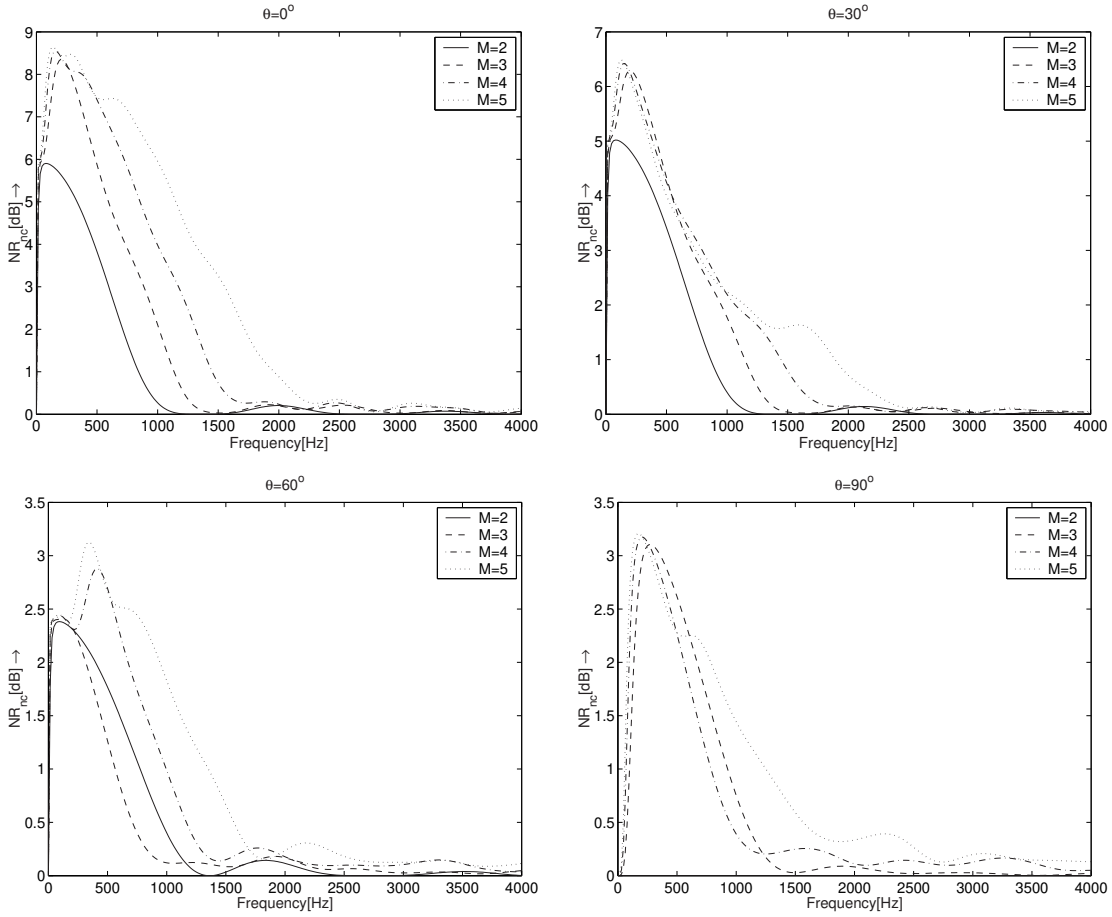


Fig. 7. Noise reduction of noise cancelling branch for diffused noise field as a function of the steering angle and the number of sensors.

the high frequency band and only relatively low noise reduction in the low frequency band.

A.3 Incoherent Noise Field

No extra noise reduction is achieved by the noise canceller branch. The FBF branch becomes a simple *delay and Sum* array, thus the expected noise reduction is M, the number of sensors.

B. Signal's TFs: Arbitrary

The more general case where arbitrary TFs relate the sources and the sensors is more complicated to introduce, as the expected performance depends on the actual TFs used.

For the evaluation of the general TFs case, we used acoustical TFs (ATFs) presented before. Three signal sources were used and received by the microphone array shown in Figure 3. The first source is located to the left of the array. The second source is located at the same source position, but with barrier placed between the source and the array. The third source is located to the right of the array. We analyze now the dependence on the noise field.

B.1 Noise Field: Arbitrary TFs

The algorithm was designed (i.e., $\boldsymbol{H}(e^{j\omega})$ and $\boldsymbol{G}(t, e^{j\omega})$ were determined) to receive signals from the left direction and to cancel noise from the right direction. Results for the three figures of merit introduced before are presented: normalized FBF output, normalized total output and extra noise reduction of the noise cancelling branch. If the TFs' ratios are known exactly, no attenuation is imposed on signals transmitted from the left direction. In Figure 8 the array response for a noise source transmitted from the right direction is introduced. Noise reduction of about 70 dB is demonstrated in the interesting frequency band.

If the signal is transmitted from the barriered left direction (for an array designed to receive signals from the left direction) the amount of cancellation is far inferior, as can be seen from Figure 9. This fact demonstrates the sensitivity to errors in estimating the TFs' ratios.

B.2 Diffused Noise Field

If the noise field is diffused, the noise reduction of the noise cancelling branch of the array while designed to receive signals from the three directions is given in Figure 10 for the "left", "left with barrier" and "right" directions, respectively. It is clearly shown, that the expected performance of the algorithm in diffused field is inferior to the expected performance for point source noise field. Although a considerable noise reduction is obtained in the low frequencies, it

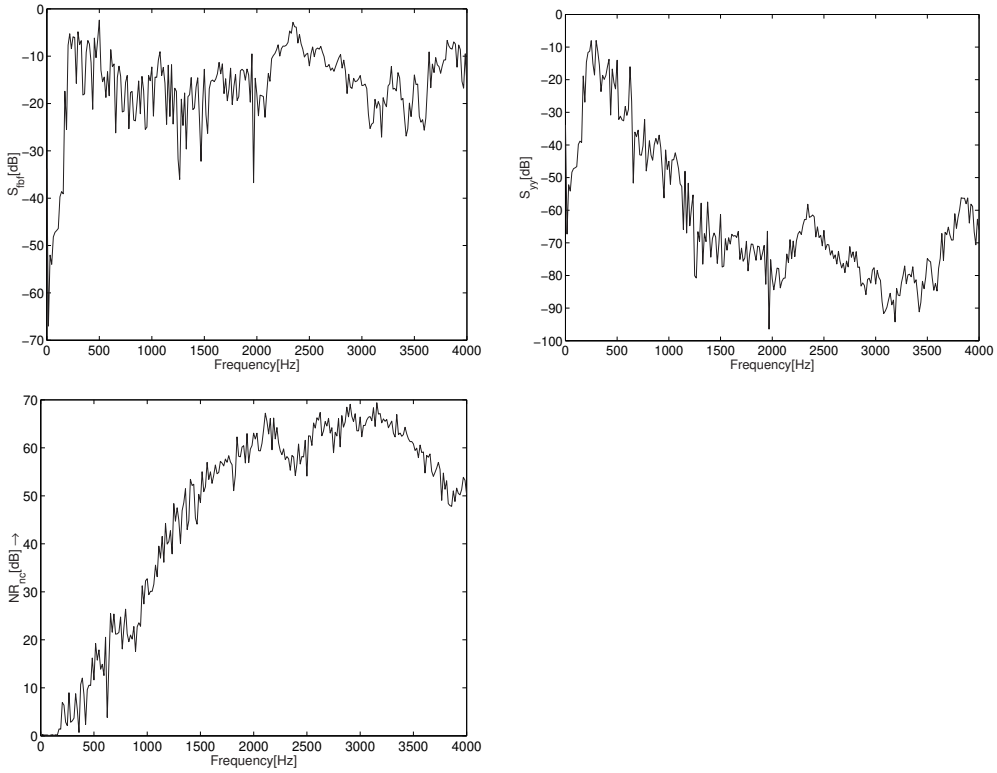


Fig. 8. Expected performance - array designed for "left" originating signals. Signal received from "right" direction.

should be emphasized that almost no signal is present in this band, as can be shown from the typical ATF introduced in Figure 3

B.3 Incoherent noise field

In an incoherent noise field, there is no extra noise reduction achieved by the noise cancelling branch. The only noise reduction is caused by the FBF branch. Almost no noise reduction is achieved.

VIII. Summary

We have derived a general expression for the output signal spectrum. From this expression we derived expressions for the expected distortion imposed on the desired signal and the amount of achievable noise reduction. The distortion depends on the quality of estimating the ratio of TFs. The noise reduction depends on the TFs' ratio involved and the noise field. High noise reduction is achieved in a coherent noise field. In an incoherent noise field only the FBF branch is responsible for noise reduction. The amount of noise reduction depends on the TFs'

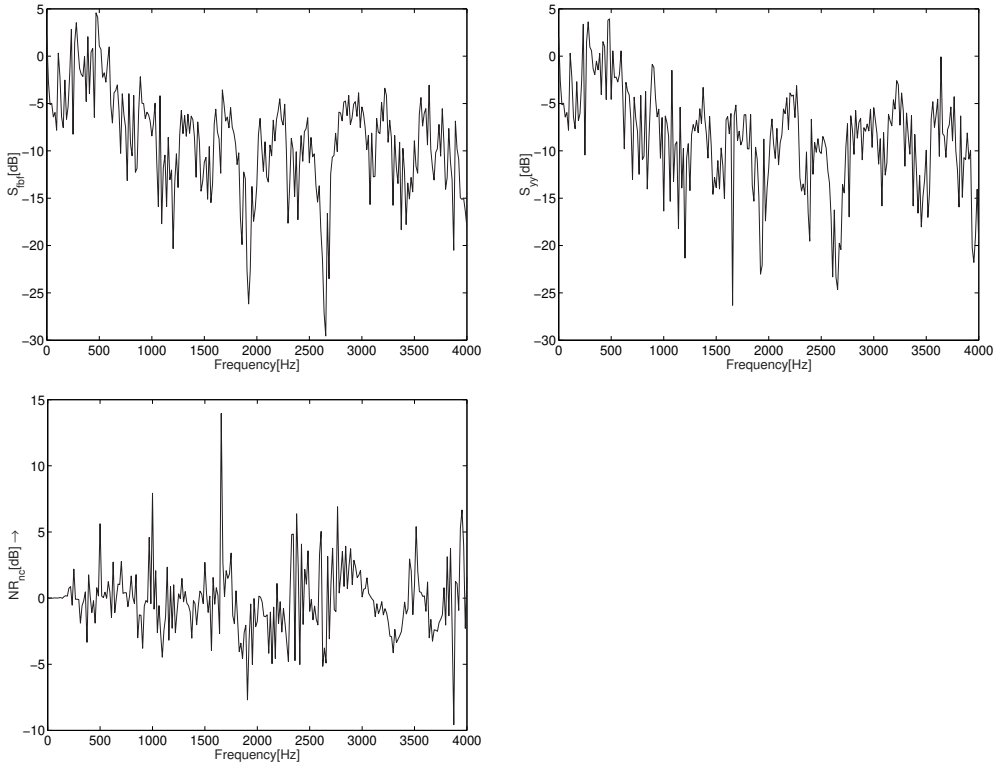


Fig. 9. Expected performance - array designed for "left" originating signals. Signal received from "left with barrier" direction.

ratio involved, and it might be negligible. In a diffused noise field we expect a significant noise reduction only in the lower frequency band, but recalling that there is almost no desired signal in that band, the amount of noise reduction may also be almost negligible. We note that in actual scenarios the noise field is a mixture between point sources and diffused sources, thus the expected result should fall between these two extremes. Recall that in practice the suggested algorithm uses sequential update for the filters involved rather than the closed form optimal solution. This might degrade the performance of the algorithm, yielding inferior results than predicted in this study. Nevertheless, while it is commonly known that the performance of the classical Griffiths & Jim GSC algorithm severely degrades in the general TFs case, it was shown that the recently proposed TF–GSC algorithm still maintains its good performance in terms of both noise reduction and signal distortion figures of merit. The analytical performance evaluation presented here supports the previously observed experimental evidence.

This analysis can serve as a framework for evaluating the expected performance of other GSC structures in a reverberant environments with only minor modifications.

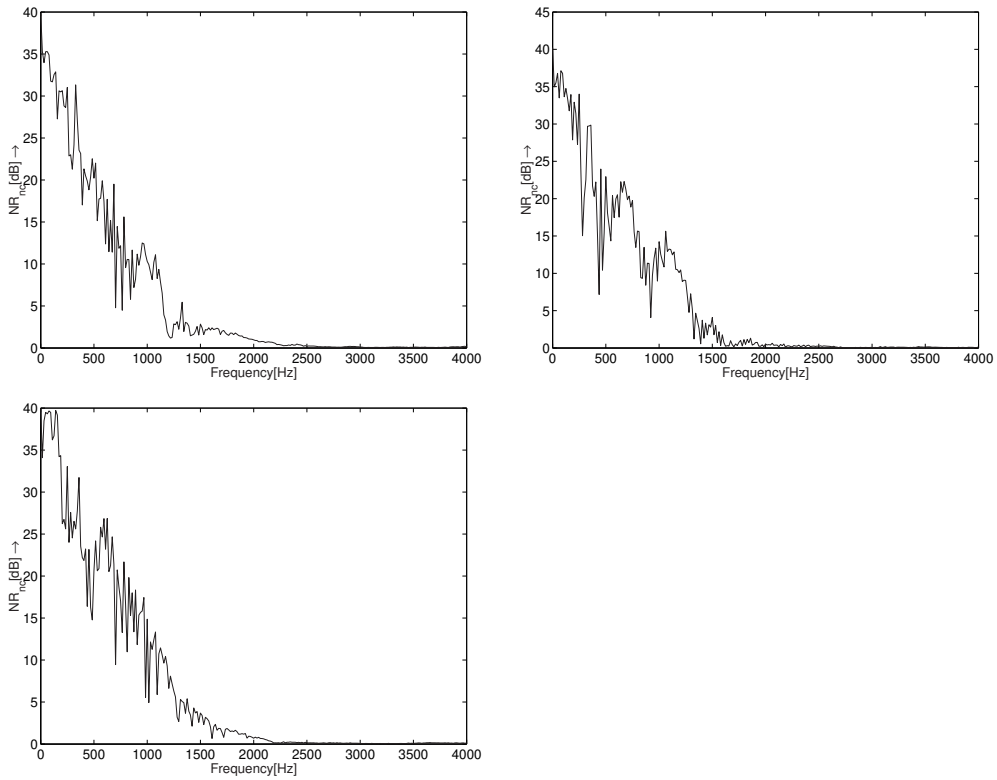


Fig. 10. Noise reduction of the noise cancelling branch designed for "left" (top-left) "left with barrier" (top-right), "right" (bottom-left) originating signals and diffused noise field.

Appendix

I. Derivation of Expected Noise Reduction for Coherent Noise Field

Recall 21 and define,

$$\mathcal{X}(t,e^{j\omega}) = \Phi_{\boldsymbol{NN}}(t,e^{j\omega})\mathcal{H}(e^{j\omega})\left(\mathcal{H}(e^{j\omega})^{\dagger}\Phi_{\boldsymbol{NN}}(t,e^{j\omega})\mathcal{H}(e^{j\omega})\right)^{-1}\mathcal{H}^{\dagger}(e^{j\omega})\Phi_{\boldsymbol{NN}}(t,e^{j\omega})$$

For clarity and simplicity we will omit the time and frequency dependence in the next derivation. Thus,

$$\mathcal{X}(t,e^{j\omega}) = \mathcal{X} = \Phi_{\boldsymbol{NN}}\mathcal{H}\left(\mathcal{H}^{\dagger}\Phi_{\boldsymbol{NN}}\mathcal{H}\right)^{-1}\mathcal{H}^{\dagger}\Phi_{\boldsymbol{NN}}.$$

We further denote,

$$\mathcal{X} = \mathcal{K}\times\mathcal{L}\times\mathcal{M}$$

where, $\mathcal{K} = \Phi_{\boldsymbol{NN}}\mathcal{H}$, $\mathcal{L} = \left(\mathcal{H}^{\dagger}\Phi_{\boldsymbol{NN}}\mathcal{H}\right)^{-1}$ and $\mathcal{M} = \mathcal{H}^{\dagger}\Phi_{\boldsymbol{NN}}$. Thus, $\mathcal{X}$ is a multiplication of three terms. Starting from $\mathcal{L}$ and using the detailed noise structure,

$$\mathcal{L} = \left(\mathcal{H}^{\dagger}(\Phi_{nn}\boldsymbol{B}\boldsymbol{B}^{\dagger}+\varepsilon I)\mathcal{H}\right)^{-1} = \left(\Phi_{nn}(\mathcal{H}^{\dagger}\boldsymbol{B})(\mathcal{H}^{\dagger}\boldsymbol{B})^{\dagger}+\varepsilon\mathcal{H}^{\dagger}\mathcal{H}\right)^{-1}.$$

If $\boldsymbol{B}(e^{j\omega}) = \boldsymbol{A}(e^{j\omega})$, i.e., the noise source is located exactly at the desired signal location, than $\mathcal{H}^{\dagger}(e^{j\omega})\boldsymbol{B}(e^{j\omega}) = 0$, and the calculation of the inverse is straight forward, yielding $\mathcal{L} = (\varepsilon\mathcal{H}^{\dagger}\mathcal{H})^{-1}$, $\mathcal{K} = \varepsilon\mathcal{H}$ and $\mathcal{M} = \varepsilon\mathcal{H}^{\dagger}$. Collecting terms we obtain, $\mathcal{X} = \frac{\varepsilon^2}{\varepsilon}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger} \overset{\varepsilon\to 0}{\longrightarrow} 0$ Thus, no extra noise reduction is produced by the noise canceller branch, as expected. The total noise part of the output is given by,

$$\Phi_{yy}^{n}(t, e^{j\omega}) = \Phi_{fbf}^{n}(t, e^{j\omega}) = |\mathcal{F}(e^{j\omega})|^2 |A_1(e^{j\omega})|^2$$

which is exactly the signal part derived in Section IV.

For the general case, $\boldsymbol{B}(e^{j\omega}) \neq \boldsymbol{A}(e^{j\omega})$, we use the *Matrix Inversion Lemma*, yielding:

$$\mathcal{L} = \left(\frac{1}{\varepsilon}(\mathcal{H}^{\dagger}\mathcal{H})^{-1} - \frac{\frac{1}{\varepsilon^2}\Phi_{nn}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}}{1 + \frac{1}{\varepsilon}\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}} \right).$$

Now, using the approximation $\frac{1}{1+\mu} \approx 1 - \mu$, for $\mu \to 0$ (μ properly defined), yields,

$$\mathcal{L} = \left(\frac{1}{\varepsilon}(\mathcal{H}^{\dagger}\mathcal{H})^{-1} - \frac{\frac{1}{\varepsilon}\Phi_{nn}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}}{\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}} + \frac{\Phi_{nn}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}}{\left(\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\right)^2} \right).$$

Now, calculating $\mathcal{X}$,

$$\mathcal{X} = \mathcal{K}\mathcal{L}\mathcal{M} = \Phi_{\boldsymbol{NN}}\mathcal{H}\mathcal{L}\mathcal{H}^{\dagger}\Phi_{\boldsymbol{NN}} =$$
$$(\Phi nn\boldsymbol{B}\boldsymbol{B}^{\dagger} + \varepsilon I)\mathcal{H}\mathcal{L}\mathcal{H}^{\dagger}(\Phi nn\boldsymbol{B}\boldsymbol{B}^{\dagger} + \varepsilon I) \overset{\triangle}{=} (\mathcal{K}_1 + \mathcal{K}_2)\mathcal{H}(\mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3)\mathcal{H}^{\dagger}(\mathcal{M}_1 + \mathcal{M}_2).$$

with the obvious definitions of $\mathcal{K}_1$, $\mathcal{K}_2$, $\mathcal{L}_1$, $\mathcal{L}_2$, $\mathcal{L}_3$, $\mathcal{M}_1$, $\mathcal{M}_2$. Opening the brackets we have twelve terms:

I. $\mathcal{K}_1\mathcal{L}_1\mathcal{M}_1 = \frac{1}{\varepsilon}\Phi_{nn}^2\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}$

II. $\mathcal{K}_1\mathcal{L}_2\mathcal{M}_1 = -\frac{1}{\varepsilon}\Phi_{nn}^3 \times \frac{\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\boldsymbol{B}\boldsymbol{B}^{\dagger}}{\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}} = -\frac{1}{\varepsilon}\Phi_{nn}^2\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}$

III. $\mathcal{K}_1\mathcal{L}_3\mathcal{M}_1 = \frac{\Phi_{nn}^3\boldsymbol{B}(\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B})^2\boldsymbol{B}^{\dagger}}{(\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B})^2} = \Phi_{nn}\boldsymbol{B}\boldsymbol{B}^{\dagger}$

IV. $\mathcal{K}_2\mathcal{L}_1\mathcal{M}_1 = \Phi_{nn}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}$

V. $\mathcal{K}_2\mathcal{L}_2\mathcal{M}_1 = -\Phi_{nn}^2\frac{\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\boldsymbol{B}\boldsymbol{B}^{\dagger}}{\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}} = -\Phi_{nn}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}$

VI. $\mathcal{K}_2\mathcal{L}_3\mathcal{M}_1 = \varepsilon\frac{\Phi_{nn}^2\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}}{(\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B})^2} = \varepsilon\frac{\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}}{\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}}$

VII. $\mathcal{K}_1\mathcal{L}_1\mathcal{M}_2 = \Phi_{nn}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}$

VIII. $\mathcal{K}_1\mathcal{L}_2\mathcal{M}_2 = -\Phi_{nn}^2\frac{\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}}{\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}} = -\Phi_{nn}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}$

IX. $\mathcal{K}_1\mathcal{L}_3\mathcal{M}_2 = \varepsilon\frac{\Phi_{nn}^2\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}}{(\Phi_{nn}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B})^2} = \varepsilon\frac{\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}}{\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}}$

X. $\mathcal{K}_2\mathcal{L}_1\mathcal{M}_2 = \varepsilon\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}$

XI. $\mathcal{K}_2\mathcal{L}_2\mathcal{M}_2 = -\varepsilon\frac{\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}}{\boldsymbol{B}^{\dagger}\mathcal{H}(\mathcal{H}^{\dagger}\mathcal{H})^{-1}\mathcal{H}^{\dagger}\boldsymbol{B}}$

XII. $\mathcal{K}_2\mathcal{L}_3\mathcal{M}_2 = \varepsilon^2 \frac{\Phi_{nn}\mathcal{H}(\mathcal{H}^\dagger\mathcal{H})^{-1}\mathcal{H}^\dagger \boldsymbol{B}\boldsymbol{B}^\dagger\mathcal{H}(\mathcal{H}^\dagger\mathcal{H})^{-1}\mathcal{H}^\dagger}{(\Phi_{nn}\boldsymbol{B}^\dagger\mathcal{H}(\mathcal{H}^\dagger\mathcal{H})^{-1}\mathcal{H}^\dagger\boldsymbol{B})^2}$

Note, that terms I,II terms IV,V and terms VII,VIII eliminate each other, and that terms VI,IX,X,XI,XII tend to zero as ε tends to zero. Only term III is left. Collecting all terms, we have for the noise part of the output,

$$\Phi^n_{yy}(t, e^{j\omega}) = \frac{|\mathcal{F}(e^{j\omega})|^2}{\|\boldsymbol{H}(e^{j\omega})\|^4} \times \left\{ \boldsymbol{H}^\dagger(e^{j\omega})\Phi_{\boldsymbol{NN}}(t, e^{j\omega})\boldsymbol{H}(e^{j\omega}) - \boldsymbol{H}^\dagger(e^{j\omega})\Phi_{nn}(t, e^{j\omega})\boldsymbol{B}(e^{j\omega})\boldsymbol{B}^\dagger(e^{j\omega})\boldsymbol{H}(e^{j\omega}) \right\} = 0$$

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